

# Exponential Expressions

*Teacher's Edition*

## The Period

**MA.7.NSO.1.1** Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational-number bases.

**Must** Two or three laws in one expression, whole-number exponents and rational bases; generate an equivalent expression and evaluate it.

**Must not** No negative exponents yet. The laws never cross addition or subtraction, and order of operations stays at six steps or fewer.

**Target** I can apply more than one law of exponents to evaluate an expression and to write an equivalent one.

**MTR.2.1** — Notes II and board 9 · **MTR.3.1** — Boards 1 through 4 ·

**MTR.6.1** — Board 5

| Segment                             | Slides | Time          |
|-------------------------------------|--------|---------------|
| Settle in; post the learning target | 1      | 1 min         |
| Warm-Up                             | 2-3    | 5 min         |
| Notes                               | 4-6    | 10 min        |
| Example 1                           | 7-8    | 3 min         |
| Your Turn                           | 9-10   | 3 min         |
| Example 2                           | 11-12  | 3 min         |
| Your Turn                           | 13-14  | 3 min         |
| Whiteboards                         | 15-32  | 14 min        |
| Independent Set                     | 33     | 6 min         |
| IXL                                 | 34     | 5 min         |
| <b>Total</b>                        |        | <b>53 min</b> |

### Say these three things out loud today

1. Make the plan before you make a mark. Inside the parentheses first, or expand first — either, but pick one.
2. A zero exponent is a 1 sitting in the expression. It changes nothing, and it is not nothing.
3. The laws work on multiplication and division only. They never cross a plus sign.

# Slide by Slide

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1. **Exponential Expressions.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is the prerequisite: exponents before multiplication before addition.
3. **Warm-Up — Answers.** Reveal. 1. 100 2. 7 3.  $x = 5$  4. 50
4. **Notes — Copy the table. It is your reference for the rest of the unit.** This table is the durable record: students have no worksheet and no study guide, so the notes are what they keep (ruling 11).
5. **Notes — Two students, two orders, one value.** Pick the order that makes the numbers smallest soonest. Inside the parentheses first is usually that order.
6. **Notes.** The laws are shortcuts for multiplying and dividing powers.
7. **Example 1 — A zero exponent inside a product inside a quotient.** SPOKEN ONLY.
8. **Example 1 — Worked — Caroline's plan and Jamir's plan.** Caroline adds  $5 + 0$  first (product of powers), then subtracts.
9. **Your Turn — Same steps, your numbers. Boards up when done.** Catches a student who reads  $5^0$  as 0 (the whole product becomes 0) and one who adds  $4 + 0 + 2$  ( $5^6$ ).
10. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
11. **Example 2 — Three laws, a decimal and a negative base.** SPOKEN ONLY.
12. **Example 2 — Worked.** Zero exponent, then power of a quotient backwards, then one power of  $-10$ .
13. **Your Turn — Same steps, your numbers. Boards up when done.** Catches a dropped sign (27) and a student who applied the zero law to the wrong power ( $7 \cdot$  something).
14. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
- 15-16. **Board 1.** A warehouse holds  $6^4$  boxes in  $6^2$  equal rows. Each box weighs 3 kilograms. How many boxes are in each row? → **36** *The weight of 3 kilograms is not needed. 1296 — the  $6^2$  rows were ignored [the quotient was never taken];  $6^6$  or 46,656 — the exponents were added [confused with product of powers]; anything built from 3 — the unneeded figure was used*
- 17-18. **Board 2.**  $(3^0 \cdot 3^2)^3$  → **729** *243 — added the exponents  $2 + 3$  in the power of a power step [confused with product of powers]; 0 — zero exponent read as 0*
- 19-20. **Board 3.**  $\frac{(0.5^3)^2}{(0.5)^2}$  → **0.0625** *0.125 — divided the exponents  $6 \div 2$  [7.NSO.1.1 misconception: 'divide' applied to the exponents]; 0.015625 —  $0.5^6$ , the denominator ignored*
- 21-22. **Board 4.**  $\left(\frac{2.5^5}{2.5^3}\right)^1$  → **6.25** *2.5 — the outside 1 was treated as the final exponent [identity law misapplied]; 15.625 — used  $2.5^3$*
- 23-24. **Board 5.**  $\frac{(2^2 \cdot 3.2^0)^6}{2^{10}}$  Which is the value? → **D — 4** *(A)  $(2^2)^6$  written as  $2^8$  by adding  $2 + 6$ , then  $2^8 \div 2^{10}$  [power of a power confused with product of powers]; (B) added the exponents  $12 + 10$  instead of subtracting [confused with product of powers]; (C)  $3.2^0$*

read as 0, so the whole numerator is 0 [8.NSO.1.3: zero exponent law,  $a^0 = 1$ ] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

**25-26. Board 6.**  $\frac{0.2^3}{2^3} \cdot 0.1^2$  Which expression is equivalent? → **C —  $0.1^5$**  (A) quotient of powers applied to DIFFERENT bases, 0.2 and 2 [7.NSO.1.1: the quotient law needs the same base]; (B) the  $2^3$  in the denominator was ignored [order of operations: the fraction bar is a grouping]; (D) multiplied the exponents  $3 \cdot 2$  [confused with power of a power] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

**27-28. Board 7.**  $0.1^2 \cdot (0.1^1)^2$  → **0.0001**  $0.001 = 0.1^3$ , the outer square on  $(0.1^1)^2$  dropped, leaving  $0.1^2 \cdot 0.1^1$  [power of a power step skipped];  $0.01 = 0.1^2$

**29-30. Board 8.**  $\frac{(-4)^2 \cdot (-4)^1}{(-4)^0}$  → **-64** 64 — dropped the sign; 'undefined' —  $(-4)^0$  in the denominator read as 0, so a division by 0 [8.NSO.1.3: zero exponent law,  $a^0 = 1$ ]

**31-32. Board 9.** Winston writes  $(4^7/4^4)^2 = (4^3)^2$ . Yvette writes  $(4^7/4^4)^2 = 4^{14}/4^8$ . Do they get the same value? Explain. → **Yes — both give  $4^6 = 4,096$**  'No' — judged equivalence by appearance instead of value [7.NSO.1.1: generate equivalent expressions]

**33. Independent Set — Six questions. On your own, in writing.** Hand out the Independent Set; the same six questions are on the board.

**34. IXL — Last five minutes.** IXL: every listed skill is required, SmartScore 67.

## Misconceptions to Watch

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**Board 1.** 1296 — the  $6^2$  rows were ignored [the quotient was never taken]

**Board 2.** 243 — added the exponents  $2 + 3$  in the power of a power step [confused with product of powers]

**Board 3.** 0.125 — divided the exponents  $6 \div 2$  [7.NSO.1.1 misconception: 'divide' applied to the exponents]

**Board 4.** 2.5 — the outside 1 was treated as the final exponent [identity law misapplied]

**Board 5.**  $(2^2)^6$  written as  $2^8$  by adding  $2 + 6$ , then  $2^8 \div 2^{10}$  [power of a power confused with product of powers]

**Board 6.** quotient of powers applied to DIFFERENT bases, 0.2 and 2 [7.NSO.1.1: the quotient law needs the same base]

**Board 7.**  $0.001 = 0.1^3$ , the outer square on  $(0.1^1)^2$  dropped, leaving  $0.1^2 \cdot 0.1^1$  [power of a power step skipped]

**Board 8.** 64 — dropped the sign

**Board 9.** 'No' — judged equivalence by appearance instead of value [7.NSO.1.1: generate equivalent expressions]

**Materials** Whiteboards and markers. Calculators are allowed on everything; the decimal values in the bank (5.0625, 2,293.45007, 9,605.9601) are calculator work once the single power is found.